

Design and Simulation of a Vivaldi Antenna in CST Microwave Studio

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Presentation Overview

- 1 Introduction
- 2 Design Methodology
- 3 Simulation Results
- 4 Conclusion

What is a Vivaldi Antenna?

- A **Vivaldi antenna** (or Exponentially Tapered Slot Antenna, ETSA) is a type of **broadband end-fire radiator**.
- It consists of an exponentially flared slotline etched on a dielectric substrate.
- The structure supports traveling-wave propagation that gradually radiates energy as the slot widens.
- Key features:
 - Very wide impedance bandwidth
 - High directivity
 - Planar, lightweight, and easily fabricated

Applications of Vivaldi Antennas

- **Millimeter-Wave Communication:** High-speed 5G/6G links and on-chip communication.
- **Radar Systems:** Automotive, imaging, and synthetic aperture radars.
- **Remote Sensing:** Ground-penetrating and through-wall imaging.
- **Antenna Arrays:** Used in phased arrays for beam steering and wideband coverage.
- **Scientific Instrumentation:** Radio astronomy and mmWave measurement systems.

Objectives

- Design and simulate a broadband Vivaldi antenna tested from **30–100 GHz**.
- Optimize the structure for **efficient operation between 40–60 GHz**.
- Implement an **exponential taper geometry** for wideband radiation.
- Integrate a **microstrip-to-slotline transition** for broadband feeding.
- Evaluate:
 - S_{11} characteristics
 - Radiation and total efficiency
 - Far-field radiation patterns

Exponential Taper Equation:

$$y(x) = se^{rx}$$

- $s = 0.05$ mm, $r = 1.1$
- Defines the aperture flare of the slot.

Cutoff Frequency:

$$f_c = \frac{c}{2\pi a \sqrt{\epsilon_{\text{eff}}}}$$

- $k_c = 4.33$ ensures radiation begins efficiently above the cutoff.

Antenna Schematic

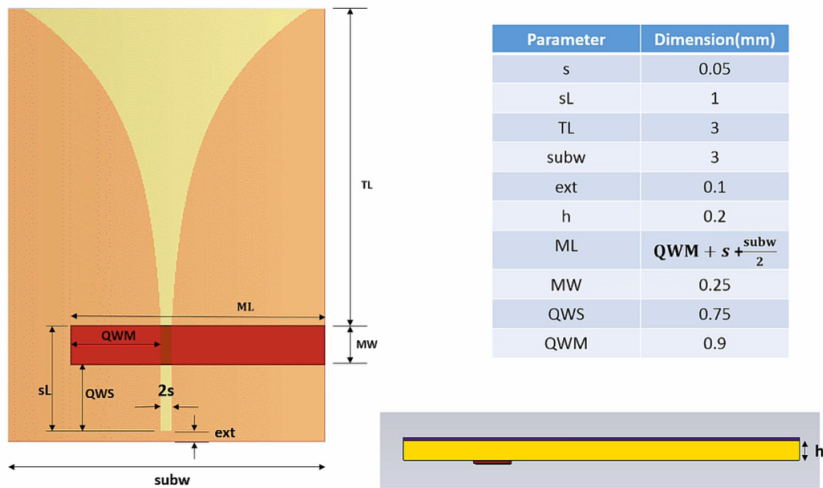


Figure: Antenna schematics and parameters.

Microstrip-to-Slotline Transition

- Converts the microstrip's quasi-TEM mode into the slotline's odd mode.
- Designed for a **50** microstrip line.
- Substrate: Rogers RO4003C ($\epsilon_r = 3.6$, $h = 0.2$ mm).
- The port geometry defines the excitation field region.

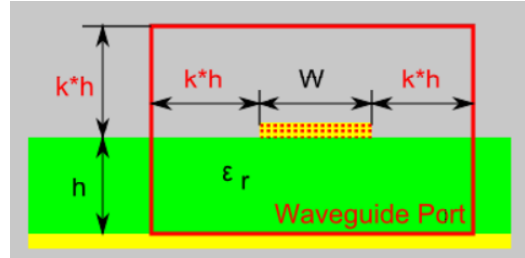


Figure: Waveguide port configuration in CST.

Antenna Design in CST

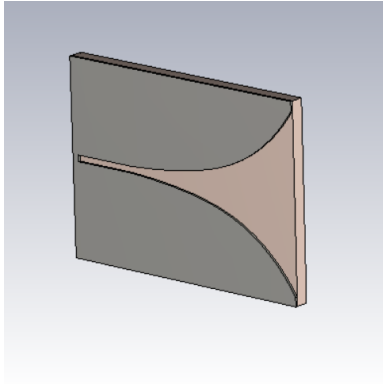


Figure: Front view.

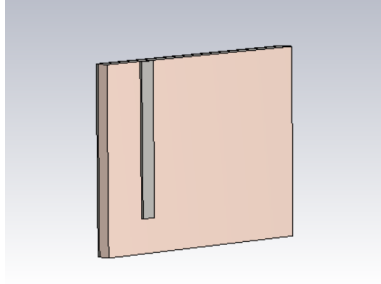


Figure: Back view.

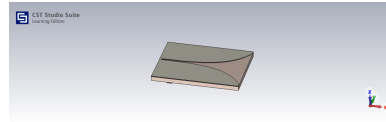


Figure: Isometric view.

Material Parameters

Parameter	Symbol	Value
Substrate material		Rogers RO4003C
Substrate thickness	h	0.2 mm
Relative permittivity	ϵ_r	3.6
Loss tangent	$\tan \delta$	0.0027
Conductor		Copper (0.035 mm)
Conductivity		5.8×10^7 S/m
Cutoff coefficient	k_c	4.33

Reflection Coefficient (S_{11})

- Minimum $S_{11} = -21.37$ dB at 46.73 GHz.
- Operating band: **40–60 GHz**.
- Impedance bandwidth (for $S_{11} < -10$ dB): approx. 38–70 GHz.

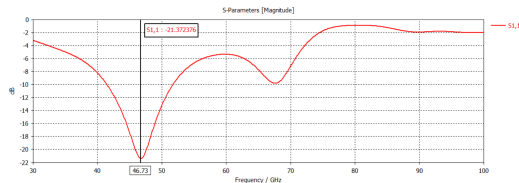


Figure: Simulated S_{11} vs Frequency.

Radiation and Total Efficiency

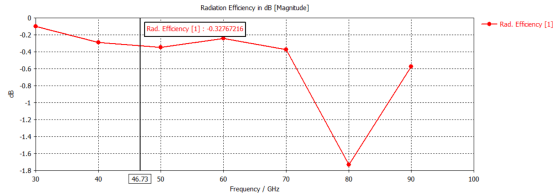


Figure: Radiation efficiency vs frequency.

- Radiation efficiency -0.33 dB at 46 GHz (93
- Total efficiency -0.55 dB at 50 GHz (88

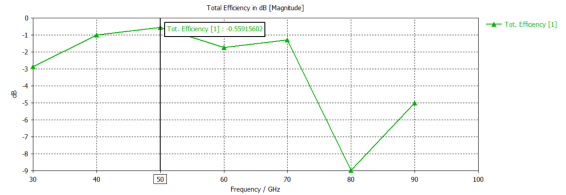
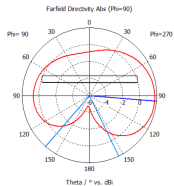


Figure: Total efficiency vs frequency.

Radiation Patterns at 40 GHz



Frequency = 40 GHz
Main lobe magnitude = 1.77 dB
Main lobe direction = 95.0 deg.
Angular width (3 dB) = 293.3 deg.

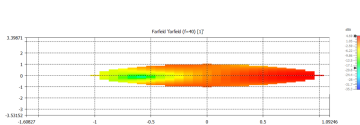


Figure: 2D pattern.

Figure: 1D Polar plot.

- End-fire radiation with a main lobe at 95° .
- Main lobe magnitude = 1.77 dBi; 3 dB width 293° .

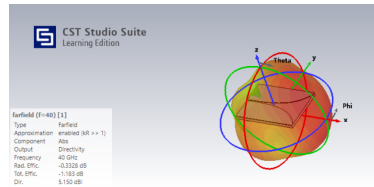


Figure: 3D pattern (Directivity = 4.69 dBi).

Performance Summary

Parameter	Frequency (GHz)	Value
Min S_{11}	46.73	-21.37 dB
Directivity	40	4.69 dBi
Radiation Efficiency	46	-0.33 dB
Total Efficiency	50	-0.55 dB
Operating Band		40–60 GHz
Test Range		30–100 GHz

- The designed Vivaldi antenna demonstrates wideband performance tested from 30–100 GHz.
- Efficient radiation observed across the 40–60 GHz band.
- Achieved:
 - $S_{11} = -21.37$ dB at 46.73 GHz
 - Radiation efficiency 93%
 - Total efficiency 88%
 - Directivity 4.69 dBi
- Suitable for mmWave communication and imaging applications.

References



Md Masud Rana, Reshma Khanom, and Md Mostafizur Rahman, "Design and Analysis of Vivaldi Antennas," *Proc. 2018 Int. Conf. on Innovation in Engineering and Technology (ICIET)*, pp. 1–5, 2018.



CST Microwave Studio, Dassault Systèmes, 2025. <https://www.3ds.com/products/simulia/cst-studio-suite>